TRILBY A. HILL

MECHANICAL ENGINEERING STUDENT

trilbyadara@gmail.com | (985) 773 9504 | PO Box 766 Sun, LA 70463

Education

Louisiana Tech University, Ruston, LA
College of Engineering and Science
BS Mechanical Engineering
Overall GPA: 3.41/4.0
Expected Graduation: May 2026

- Louisiana Tech University Society of Women Engineers (SWE), Fall 2022 - Present
  SWE Public Relations Chair, Spring 2023 - Spring 2024
- Louisiana Tech University Pi Tau Sigma (Tau Xi Chapter) Mechanical Engineering Fraternity, Winter 2024 - Present
  Pi Tau Sigma Secretary, Spring 2025 - Present
- Louisiana Tech University Honors College, Fall 2022 - Present
  Accepted into Honors based on academic merit and 31 on ACT

Experience

Science, Mathematics, and Research for Transformation (SMART) Program | Undergraduate Intern
U.S. Department of Defense, Washington, DC **August 2024 - Present**

- Designing data-logging devices with fluid-dynamic considerations to monitor atmospheric and oceanic conditions during naval operations using SolidWorks, C++, and Arduino IDE
- Calibrating and testing data-logging devices in extreme conditions, including 30-meter ocean depths, using Python (Pandas, NumPy, Matplotlib)
- Co-Filing for a patent on innovative data-logging devices with my mentor, Dr. Matthew Hornbach

STEM Student Employment Program (SSEP) | Undergraduate Mechanical Engineering Apprentice
U.S. Naval Research Laboratory, Stennis Space Center, MS **May 2023 - August 2024**

- Conducted data mining to extract valuable environmental marine data from various regional databases
- Aided in the optimization of predicting properties in areas with sparse data via machine learning
- Scripted Python programs and operated Unix command line tools to efficiently process and analyze data

Science and Engineering Apprenticeship Program (SEAP) | High School Student Intern
U.S. Naval Research Laboratory, Stennis Space Center, MS **June – July 2021 & June – July 2022**

- Cleaned and analyzed clay mineral smear slide data collected from the seabed
- Researched empirical relationships between various properties of the seabed on a global scale
- Created posters focusing on my research, that were presented at the American Geophysical Union's Fall Meetings of 2021 & 2022

Tools:

- Python and C++ Coding
- Arduino IDE
- Microsoft Office Specialist (MOS)
- ArcGIS and QGIS
- Autodesk Certified User (ACU)
- Certified SolidWorks Associate (CSWA)
- Operation of soldering machines, milling machines, and 3D printers

Achievements and Publications:

- Named as second author on the manuscript: "A machine learning approach using legacy geophysical datasets to model Quaternary marine paleo topography" in Elsevier's *Applied Computing and Geoscience Journal*
- Awarded the "DaVinci Award" at Louisiana Tech University's annual Freshman Design Expo in May 2023
- Selected as an Emmy Noether's Women in STEM Scholar in May 2022
- Presented published research posters at the American Geophysical Union Fall Meetings of 2021 - 2024

References:

Dr. Matthew Hornbach, Geophysicist at U.S. Naval Research Laboratory- matthew.j.hornbach.civ@us.navy.mil
Dr. Blake Landry, Engineer at U.S. Naval Research Laboratory- blake.j.landry4.civ@us.navy.mil

RILEY C. LAMONTE

985.774.7668 | rileylamonte@gmail.com | linkedin.com/in/rileylamonte

EDUCATION:

Louisiana Tech University, Ruston, LA,
Sept. 2022 – Present (Expected May 2026)
B.S. Mechanical Engineering
Honors College | GPA: 3.58/4.00
Passed FE Exam, June 2025 | EIT Pending

Honors Thesis:
Machine Learning Defect Detection in Additive Manufacturing (Defense Expected March 2026)

LEADERSHIP & INVOLVEMENT:

Tau Beta Pi – Louisiana Gamma, Jan. 2025 – Present
President, Feb. 2025 – Present

Pi Tau Sigma – Tau Xi, Feb. 2025 – Present
Vice President, Mar. 2025 - Present

American Society of Mechanical Engineers – Louisiana Tech, Sept. 2022 – Present
Vice President, May 2024 – May 2025
Industry Chair, May 2023 – May 2024

EXPERIENCE:

U.S. Naval Research Laboratory (NRL) Summers 2022 – 2025
Student Employee – Pathways Internship Program *Stennis Space Center - Bay St. Louis, MS*

- Supported development of the Next-generation Educational cross-platform for Uncrewed Systems (NEXUS) platform, a sandbox software system for multi-agent unmanned vehicle coordination, hardware testing, and algorithm development.
- Designed and 3D printed robotic platforms and protective enclosures for sensor payloads using FDM additive manufacturing.
- Authored a detailed user engagement primer integrating sensor challenges, mission scenarios, and troubleshooting protocols.
- Represented NRL in STEM outreach at Infinity Science Center and naval collaboration at Oceans in Action 2025 conference.

Apprentice – Science & Engineering Apprenticeship Program (SEAP) Summer 2021 Program

- Built and programmed three remote-controlled robotic platforms as part of early development for the NEXUS system.
- Authored a LaTeX-based user manual detailing component assembly, wiring, and Python programming for reproducibility.

Moore Research Group January 2023 – Present
Undergraduate Research Assistant *Louisiana Tech University – Ruston, LA*

- Developing a real-time defect detection method for FDM-based additive manufacturing using machine learning algorithms.
- Collected and analyzed thermal data from standard and modified G-code prints to identify temperature-based anomalies.
- Conducted thermal testing on shape memory polymers (SMPs) to design a heat-activated deformation-based thermal switch.

TECHNICAL SKILLS:

Engineering Tools: Additive Manufacturing (FDM), Soldering, Milling
Design & Analysis: SolidWorks (CSWA), ANSYS, FEA, CFD, GD&T
Programming & Data: Python, MATLAB, Arduino IDE, WinDaq, Picolog
Productivity: Office, LaTeX, Mathcad, Inkscape, Fritzing, Cura, CNCjs

ASPIRATIONS:

Pass PE Exam
Earn CSWP
Pursue M.S. Mechanical Engineering
Specialize in Thermal-Fluid Systems

RESEARCH PROJECTS:

Additive Manufacturing Defect Detection System January 2023 – Present
Undergraduate Research *Moore Research Group*

Developing a real-time defect detection system for FDM 3D printing by integrating infrared (IR) sensors with machine learning algorithms to identify thermal anomalies during layer deposition. Co-authored a manuscript detailing the methodology and results.

Thermal-Activated Switch Using Shape Memory Polymers Fall 2024
Undergraduate Research *Moore Research Group*

Designed a thermal switch prototype utilizing heat-treated shape memory polymers (SMPs) to respond to temperature stimuli. Conducted thermal cycling experiments to evaluate repeatability, actuation range, and material recovery.

REFERENCES

Dr. Blake Landry, Email: blake.j.landry4.civ@us.navy.mil, Phone: 337.326.0991
Dr. Arden Moore, Email: amoore@latech.edu, Phone: 318.257.5106

RYAN C. MULKEY

Ruston, LA | 985.789.2862 | ryanmulkey17@gmail.com

EDUCATION

LOUISIANA TECH UNIVERSITY Ruston, LA
Bachelor of Science Expected May 2026
Major in Mechanical Engineering, Cumulative GPA: 3.4/4.0

LOUISIANA TECH UNIVERSITY Ruston, LA
Master of Mechanical Engineering Expected May 2027
Major in Mechanical Engineering with a focus in Thermal Fluids, Cumulative GPA: 4.0/4.0

WORK EXPERIENCE

MOORE RESEARCH GROUP Ruston, LA
Student Researcher Sept 2023 – Present
- Developing experimental procedures and conducted tests on a thermal diode system
- Plotting and interpreted data and adjusted experimental parameters accordingly

GULF STATES ENGINEERING Gulfport, MS
Mechanical Engineering Intern May 2025 – Aug 2025
- Designed HVAC and Plumbing systems for Industrial, Commercial, and Residential projects using Revit and AutoCAD
- Designed fire sprinkler systems and analyzed hazard plans using Revit and Bluebeam

TED AND CHRIS GOLF COMPANY Slidell, LA
Golf Cart Maintenance Staff May 2020 – Aug 2023
- Catered to 50-100 customers per shift while working efficiently through team management

ACTIVITIES

AMERICAN SOCIETY OF MECHANICAL ENGINEERS Ruston, LA
Secretary, May 2023 – May 2024 | President, May 2024 – May 2025 Sept 2022 – Present
- Planned and managed events for an organization of about 100 students within Louisiana Tech's College of Engineering and Science
- Directed and coordinated nine elected officers

PI TAU SIGMA- TAU XI CHAPTER Ruston, LA
President, May 2025 - Present Jan 2025 – Present
- Managed an honor society of about 25 students within Louisiana Tech's College of Engineering and Science.

ORTHODOX CHRISTIAN FELLOWSHIP Ruston, LA
President, Dec 2022 – Jan 2025 | Secretary, Jan 2025 – Present Sept 2022 – Present
- Led a religious organization of about 25 members. Coordinated talks with the student body as well as weekly meetings

PROJECTS AND TECHNICAL SKILLS

THERMAL DIODE TESTING Sept 2023
- Tested configurations of a thermal design to optimize diodicity
- Iterated the experimental design to improve test efficiency

REVIT HOUSE DESIGN July 2025
- Designed a house in Revit with custom architectural and structural families
- Features accurate ductwork and sanitary piping as well as a properly sized AHU and water heater

CERTIFICATIONS
- Engineering Tools: E.I, AutoCAD, Revit, Navisworks, SolidWorks
- Productivity: Microsoft Word, Microsoft PowerPoint, HTML/CSS, JavaScript

REFERENCES

- Jeff Dunnam – Process Engineering Department Manager at GSE – 504.231.1925
- Dr. Arden Moore – Associate Dean of Research at Louisiana Tech University – 318.257.5106
- Chris Smith – Owner of Ted and Chris Golf Company – 985.788.6139

ELISA L. TOMLIN

Ruston, LA | *318 613 3729* | elosoya563@gmail.com

Education

Louisiana Tech University, Ruston, LA
College of Engineering and Science
BS Mechanical Engineering

Major: Mechanical Engineering
Overall GPA: 3.6/4.0
Expected Graduation: May 2026

Experience

Beta Part Time Substation Engineer: Alexandria, LA (September 2024 - Present)

- Analyzing the efficiency of substation ground grid designs with CDEGS software
- Aiding in the development of large-scale substations utilizing AutoCAD software
- Revising technical drawings and correcting redlines with AutoCAD and Revu Bluebeam

Beta Transmission Engineering Internship: Alexandria, LA (June - August 2024)

- Calculated components of transmission line forces utilizing PLS-CADD software
- Collaborated with professional engineers and technicians on various high voltage projects
- Developed Beta transmission safety clearance standards based on AEP guidelines

Louisiana Tech University Freshman Design Expo: Ruston, LA (May 2023)

- Designed a self-automated guitar tuner with electrical and mechanical components
- Created a mechanical arm that moved along a linear actuator
- Utilized Arduino IDE to control motors and SOLIDWORKS designed 3D-printed parts

Skills and Abilities

- Python, Arduino IDE, and MATLAB coding
- Certified SolidWorks Associate (CSWA)
- AutoCAD and PLS-CADD software
- CDEGS software
- Proficient in Microsoft Office Excel, Word, and PowerPoint
- Can safely operate soldering machines, milling machines, and 3D printers

Achievements & Activities

Pi Tau Sigma – Tau Xi Chapter

- Inducted into Mechanical Engineering Honors Fraternity, January 2025 – Present

Louisiana Tech University Honors College

- Member of Honors College with an ACT composite score of 31, September 2022 – Present

Society of Women Engineers (SWE)

- Active member of SWE, September 2022 – Present

Davinci Award

- Awarded the “Davinci Award” at Louisiana Tech University’s Freshman Design Expo, May 2023

References

Jacob Steelman – Substation Engineering Manager, 318.419.1318
Travis Larrabee – Transmission Engineering Manager, 540.420.4972